IV received and ensuring that any new messages always use a larger number. For unreliable protocols such as CoAP over UDP, where messages may arrive out of order, you can use the algorithm from RFC 4303 (https://datatracker.ietf.org/doc/html/rfc4303#section-3.4.3). This approach maintains a window of allowed sequence numbers between a minimum and maximum value that the recipient will accept and explicitly records which values in that range have been received. If the recipient is a cluster of servers, such as a typical cloud-hosted API, then this state must be synchronized between all servers to prevent replay attacks. Alternatively, *sticky load balancing* can be used to ensure requests from the same device are always delivered to the same server instance, shown in figure 13.5, but this can be problematic in environments where servers are frequently added or removed. Section 13.1.5 discusses an alternative approach to preventing replay attacks that can be effective to REST APIs.

> **DEFINITION** ***Sticky load balancing*** is a setting supported by most load balancers that ensures that API requests from a device or client are always delivered to the same server instance. Although this can help with stateful connections it can harm scalability and is generally discouraged.